Rosenbrock: gradient-identity kappa under increasing noise

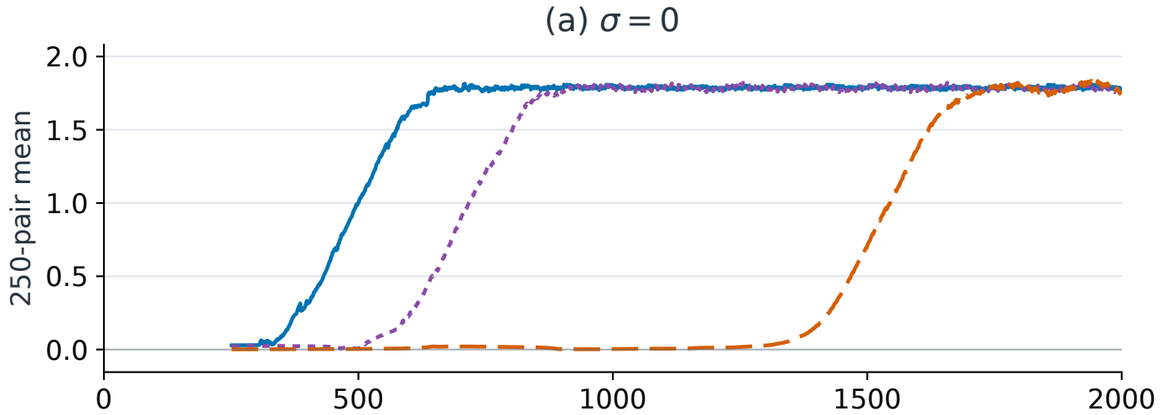
Three matched AdamW LRs | Start (0, 1) | 2,000 updates | Weight decay 0.1 | Paired noise seed 1234

$\eta = 10^{-2}$ $\eta = 3.1623 \times 10^{-3}$ $\eta = 10^{-3}$

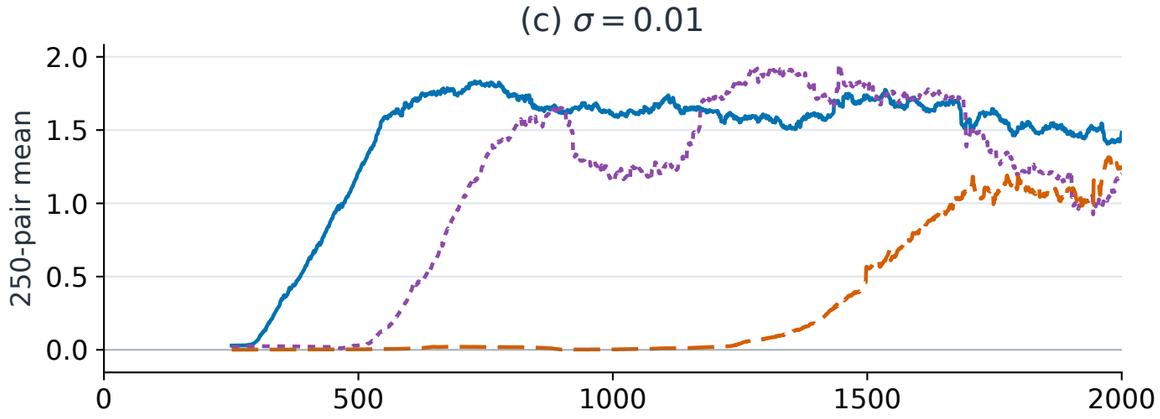
Gradient-only term: $q_t = 1 - \rho_t \|g_t\| / \|g_{t-1}\|$

Same quantity as the previous overlay; common linear Y scale

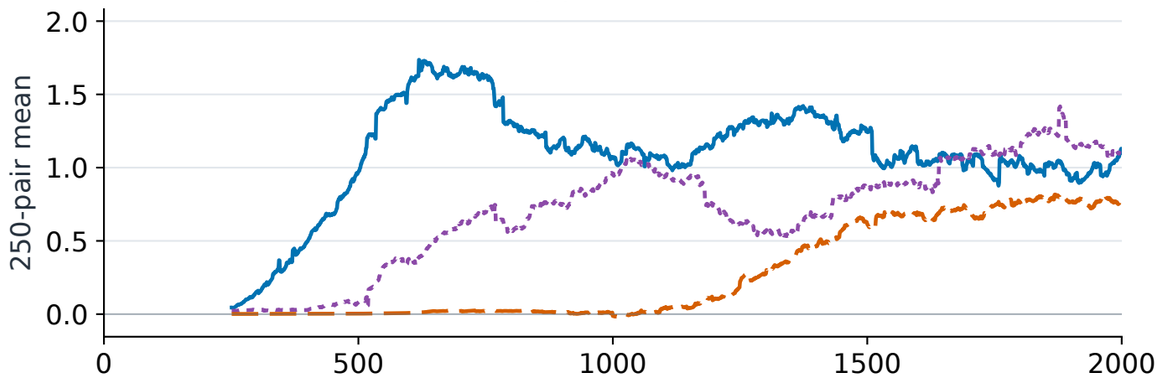
(a) $\sigma = 0$



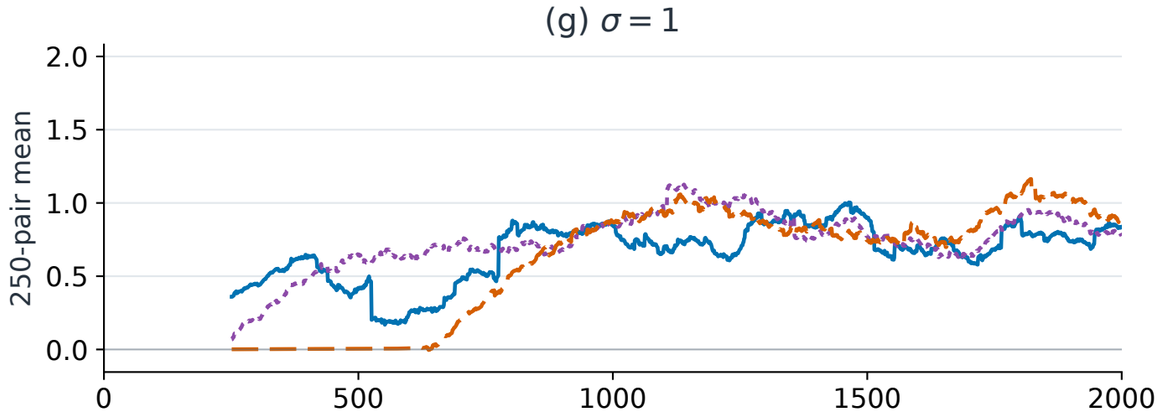
(c) $\sigma = 0.01$



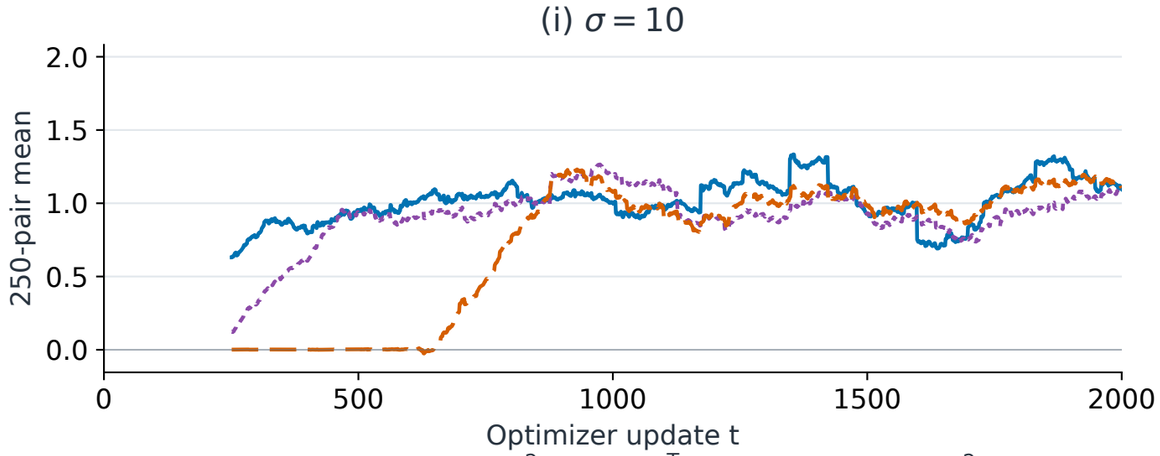
(e) $\sigma = 0.1$



(g) $\sigma = 1$



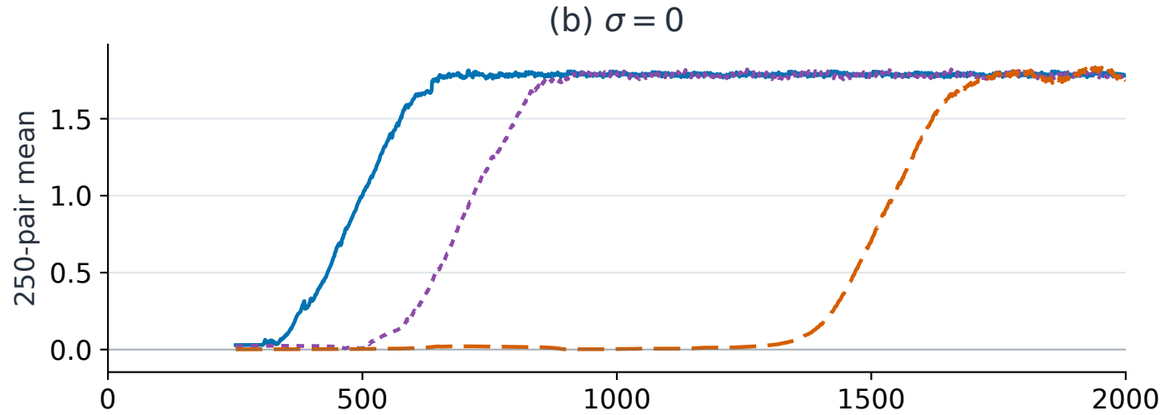
(i) $\sigma = 10$



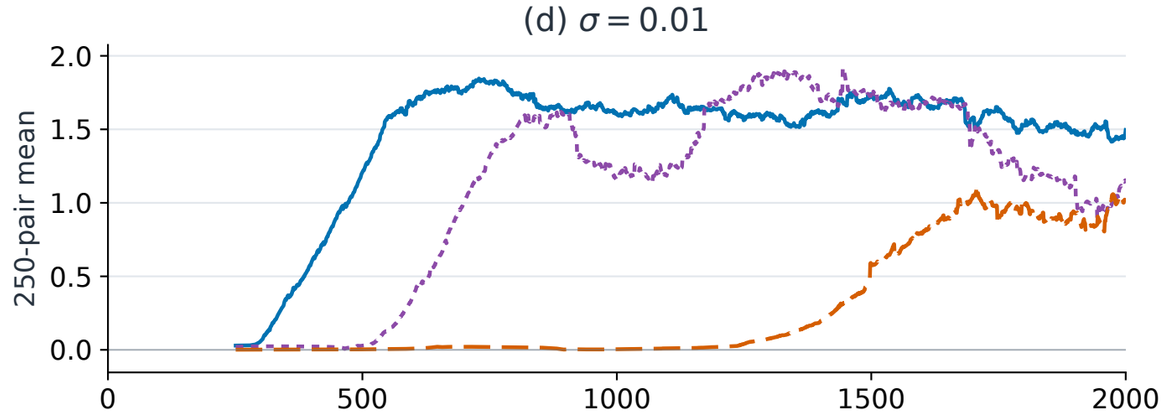
Including known noise: $\eta\hat{\kappa}_t = q_t + \varepsilon_t$

Realized noise correction; linear Y range varies by row

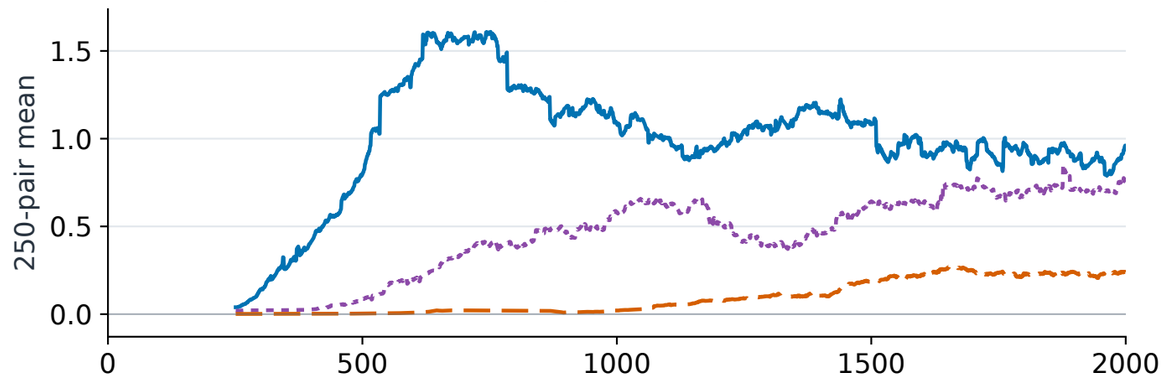
(b) $\sigma = 0$



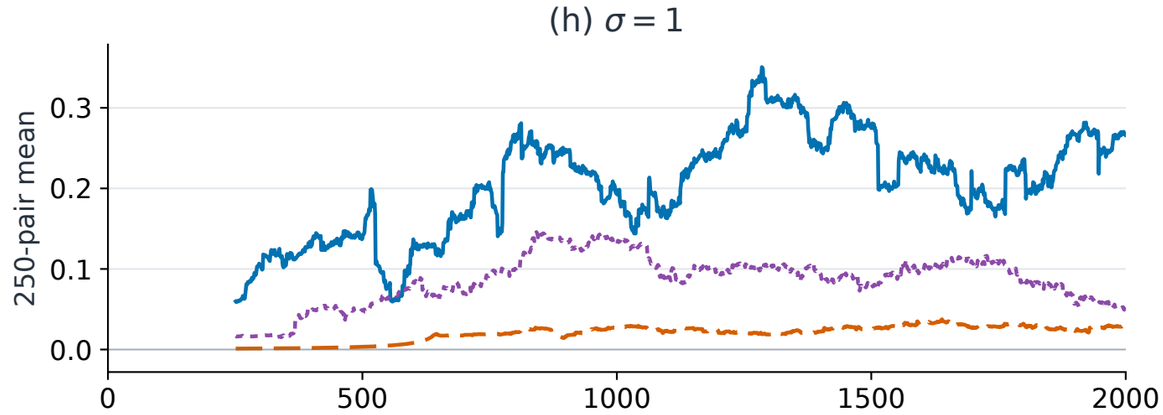
(d) $\sigma = 0.01$



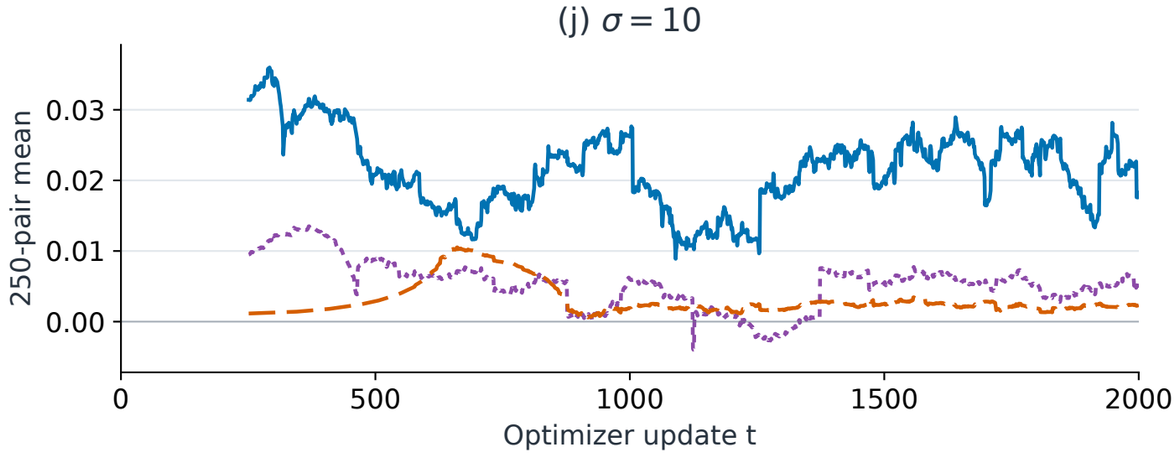
(f) $\sigma = 0.1$



(h) $\sigma = 1$



(j) $\sigma = 10$



$g_t = \nabla f(\theta_{t-1}) + \xi_t, \;\; \xi_t \sim \mathcal{N}(0, \sigma^2 I); \;\; \varepsilon_t = g_{t-1}^{\mathsf{T}}(\xi_t - \xi_{t-1}) / \|g_{t-1}\|^2.$

Compute each adjacent-pair quantity first, then average 250 pairs. First complete mean: t = 251; all 1,750 means per curve shown.

No clipping, denominator floor or seed averaging. Noise correction removes the explicit noise difference; AdamW step mismatch remains.